MORTGAGE ORIGINATION PLATFORM · ARCHITECTURE BLUEPRINT

Architecture Blueprint

Current-state and future-state architecture for the HSBC Mortgage Origination Platform. A full-stack system spanning applicant intake, underwriting, credit decisioning, and document management—built on Spring Boot, Next.js, and PostgreSQL.

Status	Version	Platform	Date
Draft	1.0	Java 17 · Spring Boot 3.2 · Next.js 14	March 2026

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Executive Summary

The Mortgage Origination Platform processes residential mortgage applications through a multi-stage workflow—from applicant intake and credit scoring through underwriter decisioning and document verification. This document maps the current implementation and charts a future-state architecture with authentication, event-driven processing, external integrations, and enterprise-grade observability.

A

Current State

Monolithic three-tier architecture. Next.js SPA, Spring Boot REST API, PostgreSQL. Mock credit checks. No auth. Docker Compose deployment.

B

Future State

Event-driven microservices. OAuth2/OIDC auth, real credit bureau integration, object storage, notification service, API gateway, and full observability.

C

Migration Path

Incremental evolution. Auth and API gateway first, then extract services behind events. No big-bang rewrite—each step delivers standalone value.

Current-State Architecture

A three-tier monolithic application deployed via Docker Compose. The frontend communicates with a single Spring Boot API backed by PostgreSQL. All processing is synchronous and in-process.

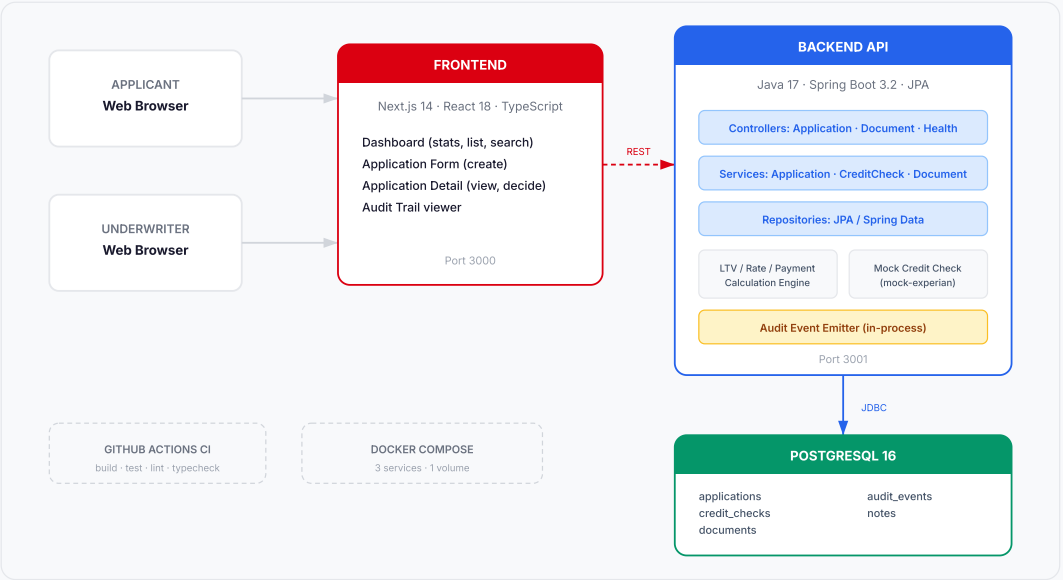

Figure 1 — Current-state system architecture. All components run as Docker containers orchestrated by Docker Compose.

Component Inventory

LAYER	TECHNOLOGY	RESPONSIBILITY	PORT
FRONTEND	Next.js 14, React 18, TypeScript	Dashboard, application form, detail view, audit trail viewer	3000
BACKEND	Java 17, Spring Boot 3.2, Spring Data JPA	REST API, business logic, mock credit scoring, audit event emission	3001
DATABASE	PostgreSQL 16 Alpine	Persistent storage for applications, documents, credit checks, audit events, notes	5432

LAYER	TECHNOLOGY	RESPONSIBILITY	PORT
CI/CD	GitHub Actions (6 parallel jobs)	Backend build/test/package, frontend lint/typecheck/build	—

Application Lifecycle

1. **Draft** — Application created with applicant, property, and loan details
2. **Submitted** — Validation passes, applicant submits for review
3. **Under Review** — Underwriter reviews application and credit data
4. **Decision** — Approved, conditionally approved, or declined

API Surface

METHOD	ENDPOINT	DESCRIPTION
`GET`	`/api/v1/applications`	List with search, filter, sort, paginate
`GET`	`/api/v1/applications/stats`	Dashboard statistics
`GET`	`/api/v1/applications/:id`	Get application by ID
`POST`	`/api/v1/applications`	Create new application
`PATCH`	`/api/v1/applications/:id`	Update application
`POST`	`/api/v1/applications/:id/submit`	Submit for underwriter review
`POST`	`/api/v1/applications/:id/decide`	Record underwriter decision
`GET`	`/api/v1/applications/:id/audit`	Retrieve audit trail
`GET`	`/api/v1/applications/:appId/documents`	List documents
`POST`	`/api/v1/applications/:appId/documents`	Upload document metadata

METHOD	ENDPOINT	DESCRIPTION
`PATCH`	`/api/v1/documents/:id/verify`	Verify document

Data Model

TABLE	KEY FIELDS	PURPOSE
`applications`	applicant_*, property_*, loan_*, status, decision, ltv_ratio, interest_rate, monthly_payment	Core mortgage application record with calculated financial metrics
`credit_checks`	credit_score, risk_band, provider, request/response_payload (JSONB)	Mock credit bureau results, one per application
`documents`	document_type, file_name, storage_path, verified, verified_by	Document metadata with verification workflow
`audit_events`	entity_type, action, actor, changes (JSONB), metadata (JSONB)	Immutable audit log for every state transition
`notes`	author, content, note_type	Freeform notes attached to applications

Current Limitations

No Authentication

All endpoints are publicly accessible. No user identity, role-based access, or session management.

Mock Credit Scoring

Credit checks use a deterministic algorithm. No integration with Experian, Equifax, or TransUnion.

No File Storage

Document endpoints store metadata only. No actual file upload or object storage integration.

Synchronous Only

All processing is synchronous. No event bus, queues, or async workflows.

No Notifications

Status changes are not communicated. No email, SMS, or push notification capability.

No Observability

No structured logging, metrics, tracing, or health dashboard beyond basic /health endpoint.

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Future-State Architecture

An event-driven, microservices-oriented platform with authentication, real integrations, async processing, and enterprise observability. Each component is independently deployable and horizontally scalable.

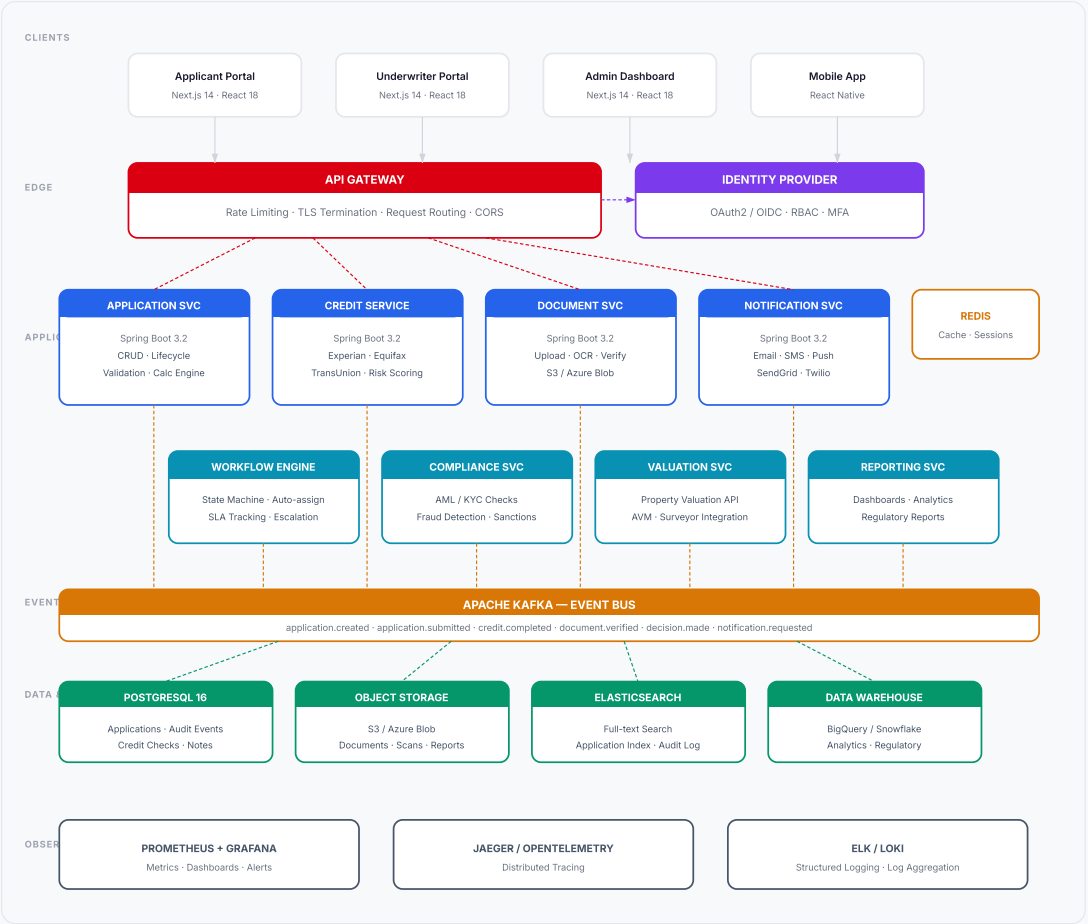

Figure 2 — Future-state target architecture. Event-driven microservices with dedicated data stores, edge security, and full observability.

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Future-State Component Detail

Detailed breakdown of each new and enhanced component in the target architecture.

COMPONENT	STATUS	TECHNOLOGY	RESPONSIBILITY
API Gateway	NEW	Kong / Spring Cloud Gateway	TLS termination, rate limiting, request routing, CORS, API versioning
Identity Provider	NEW	Keycloak / Auth0	OAuth2/OIDC, RBAC (applicant, underwriter, admin), MFA, session management
Application Service	ENHANCED	Spring Boot 3.2	Core CRUD, lifecycle management, financial calculations. Enhanced with event publishing and auth integration
Credit Service	NEW	Spring Boot 3.2	Real credit bureau integrations (Experian, Equifax, TransUnion). Multi-provider scoring with fallback
Document Service	ENHANCED	Spring Boot 3.2	File upload to S3/Azure Blob, OCR extraction, virus scanning, verification workflow
Notification Service	NEW	Spring Boot 3.2	Event-driven email/SMS/push. Templates for each lifecycle event. SendGrid + Twilio
Workflow Engine	NEW	Camunda / custom state machine	Orchestrates multi-step workflows, auto-assigns underwriters, tracks SLAs, escalation rules
Compliance Service	NEW	Spring Boot 3.2	AML/KYC checks, sanctions screening, fraud detection, regulatory rule enforcement
Valuation Service	NEW	Spring Boot 3.2	Automated Valuation Model (AVM), surveyor booking integration, property data enrichment
Reporting Service	NEW	Spring Boot 3.2	Real-time dashboards, portfolio analytics, FCA/PRA regulatory reporting

COMPONENT	STATUS	TECHNOLOGY	RESPONSIBILITY
Event Bus	NEW	Apache Kafka	Async event backbone. Topics for each domain event. Enables loose coupling between services
Redis Cache	NEW	Redis 7	Session storage, API response caching, rate limiting counters

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Current → Future Comparison

Side-by-side view of architectural decisions moving from current to future state.

CURRENT STATE

- No authentication or authorization
- Monolithic Spring Boot backend
- Synchronous request/response only
- Mock credit scoring (deterministic)
- Document metadata only (no file storage)
- No notifications
- Single PostgreSQL for all data
- No caching layer
- Basic /health endpoint only
- Docker Compose local deployment
- GitHub Actions CI (build/test/lint)
- Manual underwriter assignment

FUTURE STATE

- OAuth2/OIDC with RBAC and MFA
- Microservices behind API gateway
- Event-driven via Apache Kafka
- Real bureau integration (Experian, Equifax, TransUnion)
- S3/Azure Blob with OCR and virus scanning
- Email, SMS, push via SendGrid + Twilio
- Polyglot persistence (PG + Elasticsearch + DW)
- Redis for sessions, caching, rate limiting
- Prometheus + Grafana + Jaeger + ELK

- Kubernetes with Helm charts, multi-env
- Full CI/CD with staging, canary deploys
- Workflow engine with auto-assign + SLA

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Event-Driven Workflow

The future-state lifecycle is fully event-driven. Each state transition publishes a domain event to Kafka, triggering downstream services asynchronously.

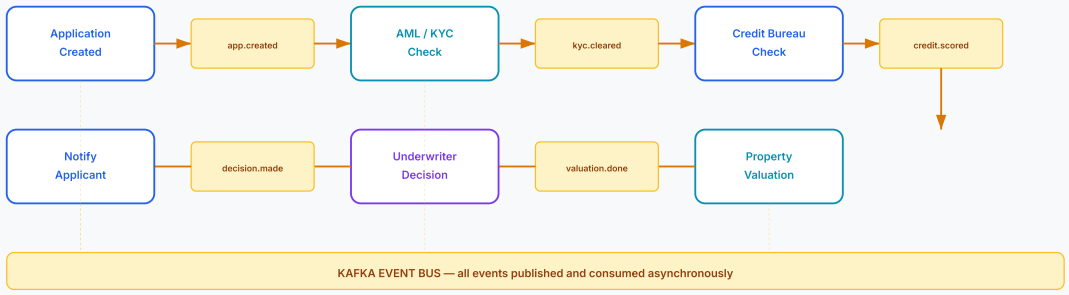

Figure 3 — Event-driven application processing pipeline. Each step publishes a domain event to Kafka, triggering the next stage.

Domain Events

EVENT	PRODUCER	CONSUMERS
`application.created`	Application Service	Compliance, Notification, Workflow
`application.submitted`	Application Service	Credit Service, Compliance, Notification
`kyc.cleared`	Compliance Service	Credit Service, Workflow
`kyc.flagged`	Compliance Service	Workflow, Notification
`credit.scored`	Credit Service	Application Service, Valuation, Workflow
`valuation.completed`	Valuation Service	Application Service, Workflow
`document.verified`	Document Service	Workflow, Notification
`decision.made`	Application Service	Notification, Reporting, Compliance
`notification.requested`	Workflow Engine	Notification Service

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Security & Access Control

Role-based access control with OAuth2/OIDC authentication, enforced at the API gateway and propagated via JWT tokens to all downstream services.

ROLE	PERMISSIONS	PORTAL ACCESS
Applicant	Create, view, update, and submit own applications. Upload documents. View own audit trail.	Applicant Portal, Mobile App
Underwriter	View all submitted/under_review applications. Run credit checks. Make decisions. Add notes.	Underwriter Portal
Admin	Full access. User management. System configuration. Reporting. Audit log export.	Admin Dashboard
System	Service-to-service authentication via client credentials. Event bus producer/consumer.	—

MFA Requirement: All underwriter and admin accounts require multi-factor authentication. Applicant accounts require MFA for document upload and application submission.

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Deployment & Infrastructure

Migration from Docker Compose to Kubernetes with multi-environment support, automated CI/CD, and infrastructure as code.

Environments

DEV	Feature branch deployments, mock integrations
STAGING	Production mirror, real integrations (sandbox), performance testing
PRODUCTION	Multi-AZ deployment, canary releases, auto-scaling

Infrastructure

Orchestration	Kubernetes (EKS / AKS)
IaC	Terraform + Helm
CI/CD	GitHub Actions → ArgoCD
Registry	ECR / ACR
DNS / CDN	CloudFront / Azure Front Door

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Implementation Roadmap

Prioritised delivery plan. Each phase is independently deployable and delivers standalone value.

P0 **Authentication & API Gateway**
OAuth2/OIDC integration, RBAC enforcement, API gateway with TLS and rate limiting

P0 **Real Credit Bureau Integration**
Extract credit service, integrate Experian/Equifax APIs with fallback and circuit breaker

P1 **Document Storage & OCR**
S3/Blob upload, virus scanning, OCR extraction for ID and payslip documents

P1 **Event Bus & Async Processing**
Kafka setup, domain event publishing, refactor audit events to async consumers

P1 **Notification Service**

Email/SMS notifications for status changes, decision outcomes, document requests

P2 **Workflow Engine**
Automated underwriter assignment, SLA tracking, escalation rules, multi-step workflows

P2 **Compliance & Fraud Detection**
AML/KYC checks, sanctions screening, fraud scoring, regulatory rule engine

P2 **Property Valuation Integration**
AVM integration, surveyor booking, property data enrichment from Land Registry

P2 **Observability Stack**
Prometheus + Grafana, OpenTelemetry tracing, structured logging with ELK/Loki

P3 **Kubernetes Migration**
Helm charts, multi-env deployment, ArgoCD GitOps, canary releases

P3 **Reporting & Analytics**
Data warehouse integration, portfolio analytics dashboards, FCA/PRA reporting

P3 **Mobile Application**
React Native app for applicant self-service, document capture, status tracking